

§5-5 The Substitution Rules

Homework : 3,4,12,13,17,21,23,36,37,43,51,65,75,79,82,84,85.

Two systematic methods to do integration :

1. The substitution rule (積分) $\Leftrightarrow$ chain rule (微分).
2. Integration by parts (積分) $\Leftrightarrow$ product rule (微分).

Example 1 : $\int 2x\sqrt{1+x^2} dx$

Key : Just like doing differentiation by chain rule, we first need to identify u.

Solution : $u = 1 + x^2$

$$\begin{aligned}\frac{du}{dx} &= 2x \Rightarrow \int \sqrt{u} du = \frac{2}{3} u^{\frac{3}{2}} + C \\ du &= 2x dx \qquad \qquad \qquad = \frac{2}{3} (1+x^2)^{\frac{3}{2}} + C.\end{aligned}$$

Example 2 : $\int \frac{x}{\sqrt{1-4x^2}} dx$

Solution : $u = 1 - 4x^2$

$$\begin{aligned}\frac{du}{dx} &= -8x \Rightarrow -\frac{1}{8} \int \frac{du}{\sqrt{u}} = -\frac{1}{8} \int u^{-\frac{1}{2}} \\ du &= -8x dx \qquad \qquad \qquad = -\frac{1}{4} u^{\frac{1}{2}} + C = -\frac{1}{4} (1-4x^2)^{\frac{1}{2}} + C \\ -\frac{1}{8} du &= x dx.\end{aligned}$$

Summary : $\int f(g(x)) g'(x) dx = \int f(u) du$

$\downarrow \qquad \downarrow$

$u \qquad du \qquad u = g(x).$

Example 3 : $\int x^3 \cos(x^4 + 2) dx$.

Solution : $u = x^4 + 2$

$$\begin{aligned} du = 4x^3 dx &\Rightarrow \int \cos u du = \frac{1}{4} \sin u + C \\ \frac{1}{4} du = x^3 dx &\Rightarrow \int \cos(x^4 + 2) dx = \frac{1}{4} \sin(x^4 + 2) + C. \end{aligned}$$

Example 4 : $\int \frac{x}{\sqrt{x-1}} dx$.

Solution : $u = x - 1 \Rightarrow x = u + 1$

$$\begin{aligned} du = dx &\Rightarrow \int \frac{(u+1) du}{\sqrt{u}} = \int \left(u^{\frac{1}{2}} + u^{-\frac{1}{2}} \right) du \\ &= \frac{2}{3} (x-1)^{\frac{3}{2}} + 2(x-1)^{\frac{1}{2}} + C. \end{aligned}$$

Example 5 : $\int \tan x dx \left(= \int \frac{\sin x}{\cos x} du \right)$.

Solution : $u = \cos x$

$$\begin{aligned} du = -\sin x dx &\Rightarrow -\int \frac{du}{u} = -\ln|u| + C \\ &= -\ln|\cos x| + C \\ &= \ln|\sec x| + C. \end{aligned}$$

Example 6 : $\int_1^e \frac{\ln x}{x} dx$.

Solution : $u = \ln x$

$$du = \frac{1}{x} dx \Rightarrow \int_0^1 u du = \frac{u^2}{2} \Big|_0^1 = \frac{1}{2}.$$

Example 7 : $\int_0^4 \sqrt{2x+1} dx$.

Solution : $u = 2x + 1$

$$\begin{aligned}
 du = 2dx & \Rightarrow \frac{1}{2} \int_1^9 u^{\frac{1}{2}} du = \frac{1}{3} u^{\frac{3}{2}} \Big|_1^9 \\
 \frac{1}{2} du = dx & \Rightarrow 9 - \frac{1}{3} = \frac{26}{3}.
 \end{aligned}$$

Example 8 : $\int \sec^3 x \tan x \, dx$.

Solution : $u = \sec x$

$$\begin{aligned}
 du = \sec x \tan x \, dx & \Rightarrow \int \sec^2 x (\sec x \tan x) \, dx \\
 & = \int u^2 \, du = \frac{u^3}{3} + C \\
 & = \frac{\sec^3 x}{3} + C.
 \end{aligned}$$

Example 9 : $\int \frac{x}{\sqrt[4]{x+2}} \, dx$.

Solution : $u = x + 2 \Rightarrow x = u - 2$

$$\begin{aligned}
 du = dx & \Rightarrow \int \frac{u-2}{u^{\frac{1}{4}}} \, du = \int u^{\frac{3}{4}} - 2u^{-\frac{1}{4}} \, du \\
 & = \frac{4}{7} u^{\frac{7}{4}} - \frac{8}{3} u^{\frac{3}{4}} + C \\
 & = \frac{4}{7} (x+2)^{\frac{7}{4}} - \frac{8}{3} (x+2)^{\frac{3}{4}} + C.
 \end{aligned}$$

Example 10 : $\int_0^4 f(x) \, dx = 10$, Find $\int_0^2 f(2x) \, dx$.

Solution : $u = 2x$

$$\begin{aligned}
 \Rightarrow \frac{1}{2} du = dx & \Rightarrow \int_0^2 f(2x) \, dx = \int_0^4 \frac{1}{2} f(u) \, du = \frac{1}{2} \int_0^4 f(u) \, du \\
 & = \frac{1}{2} \int_0^4 f(x) \, dx = 5.
 \end{aligned}$$

Example 11 : Prove that $\int_0^\pi x f(\sin x) \, dx = \frac{\pi}{2} \int_0^\pi f(\sin x) \, dx$

(Hint: $u = \pi - x$)

Solution : $u = \pi - x$

$$\begin{aligned}
\Rightarrow du &= -dx & \Rightarrow \int_0^{\pi} xf(\sin x) dx &= -\int_{\pi}^0 (\pi-u) f(\sin(\pi-u)) du \\
-du &= dx & &= \int_0^{\pi} (\pi-u) f(\sin(u)) du \\
& & &= \pi \int_0^{\pi} f(\sin(u)) du - \int_0^{\pi} uf(\sin(u)) du \\
& & &= \pi \int_0^{\pi} f(\sin(x)) dx - \int_0^{\pi} xf(\sin(x)) dx
\end{aligned}$$

That is,

$$\begin{aligned}
\int_0^{\pi} xf(\sin(x)) dx &= \pi \int_0^{\pi} f(\sin(x)) dx - \int_0^{\pi} xf(\sin(x)) dx \\
\Rightarrow \int_0^{\pi} xf(\sin(x)) dx &= \frac{\pi}{2} \int_0^{\pi} f(\sin(x)) dx.
\end{aligned}$$

Example 12 : Use example 11 to evaluate $\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx$.

Solution : Let $f(y) = \frac{y}{2 - y^2} \Rightarrow f(\sin x) = \frac{\sin x}{2 - \sin^2 x} = \frac{\sin x}{1 + \cos^2 x}$

$$\begin{aligned}
\Rightarrow \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx &\stackrel{\text{ex 11.}}{=} \frac{\pi}{2} \int_0^{\pi} \frac{\sin x}{1 + \cos^2 x} dx \\
u = \cos x &\Rightarrow -\frac{\pi}{2} \int_1^{-1} \frac{du}{1+u^2} = \frac{\pi}{2} \int_{-1}^1 \frac{du}{1+u^2} \\
-du = \sin x dx &= \frac{\pi}{2} \tan^{-1} u \Big|_{-1}^1 = \frac{\pi}{2} \left(\frac{\pi}{4} - \left(-\frac{\pi}{4} \right) \right) = \left(\frac{\pi}{2} \right)^2.
\end{aligned}$$

Symmetry :

- $f(x) = f(-x) \Leftrightarrow f : \text{even function}$
 (對稱 y 軸) $\Leftrightarrow f : \text{偶函數}$ $\Rightarrow \int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx.$
- $f(x) = -f(-x) \Leftrightarrow f : \text{odd function}$
 (對稱原點) $\Leftrightarrow f : \text{奇函數}$ $\Rightarrow \int_{-a}^a f(x) dx = 0.$